

VILDAGLIPTIN-MET MALAYSIA ALL LEAFLET

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VIDASTEAD-M 50 mg/850 mg
film-coated tablets
VIDASTEAD-M 50 mg/1000 mg
film-coated tablets

Lotus



1. NAME OF THE MEDICINAL PRODUCT

VIDASTEAD-M 50 mg/850 mg film-coated tablets
VIDASTEAD-M 50 mg/1000 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

VIDASTEAD-M 50 mg/850 mg film-coated tablets
Each film-coated tablet contains 50 mg of vildagliptin and 850 mg of metformin hydrochloride (equivalent to 660 mg of metformin).
VIDASTEAD-M 50 mg/1000 mg film-coated tablets
Each film-coated tablet contains 50 mg of vildagliptin and 1000 mg of metformin hydrochloride (equivalent to 780 mg of metformin).

For excipients, see *List of Excipients*

3. PHARMACEUTICAL FORM

Film-coated tablet.
VIDASTEAD-M 50 mg/850 mg film-coated tablets
Yellow coloured, oval shaped, biconvex, film-coated tablet, debossed with "GG2" on one side and plain on other side.
VIDASTEAD-M 50 mg/1000 mg film-coated tablets
Dark yellow coloured, oval shaped, biconvex, film-coated tablet, debossed with "GG3" on one side and plain on other side.

4. CLINICAL PARTICULARS

4.1 Indications

VIDASTEAD-M is indicated as an adjunct to diet and exercise to improve glycaemic control in adults with type 2 diabetes mellitus:

- as initial therapy when diabetes is not adequately controlled by diet and exercise alone
- in patients who are inadequately controlled with metformin hydrochloride alone
- in patients who are already being treated with the combination of vildagliptin and metformin hydrochloride, as separate tablets
- in combination with other medicinal products for the treatment of diabetes, including insulin when these do not provide adequate glycaemic control

4.2 Posology and Administration

Posology

Adults with normal renal function (GFR ≥90 mL/min)

The dose of anti-hyperglycaemic therapy with VIDASTEAD-M should be individualized on the basis of the patient's current regimen, effectiveness and tolerability while not exceeding the maximum recommended daily dose of 100 mg vildagliptin. VIDASTEAD-M may be initiated at either the 50 mg/500 mg, 50 mg/850 mg or 50 mg/1000 mg tablet strength twice daily, one tablet in the morning and the other in the evening.

- For patients inadequately controlled at their maximal tolerated dose of metformin monotherapy:
The starting dose of VIDASTEAD-M should provide vildagliptin as 50 mg twice daily (100 mg total daily dose) plus the dose of metformin already being taken.
- For patients switching from co-administration of vildagliptin and metformin as separate tablets:
VIDASTEAD-M should be initiated at the dose of vildagliptin and metformin already being taken.
- For patients inadequately controlled on dual combination with metformin and a sulphonylurea:
The doses of VIDASTEAD-M should provide vildagliptin as 50 mg twice daily (100 mg total daily dose) and a dose of metformin similar to the dose already being taken. When VIDASTEAD-M is used in combination with a sulphonylurea, a lower dose of the sulphonylurea may be considered to reduce the risk of hypoglycaemia.
- For patients inadequately controlled on dual combination therapy with insulin and the maximal tolerated dose of metformin:
The dose of VIDASTEAD-M should provide vildagliptin dosed as 50 mg twice daily (100 mg total daily dose) and a dose of metformin similar to the dose already being taken.
- For treatment naïve patients:
VIDASTEAD-M may be initiated at 50 mg/500 mg once daily and gradually titrated to a maximum dose of 50 mg/1000 mg twice daily after assessing the adequacy of therapeutic response.

The safety and efficacy of vildagliptin and metformin as triple oral therapy in combination with a thiazolidinedione have not been established.

Note: VIDASTEAD-M is available in the strength of 50 mg/850 mg and 50 mg/1000 mg. It is not able to deliver all approved dose regimens of 50 mg/500 mg. Other similar products with approved strength of 50 mg/500 mg should be used in such cases.

Special populations

Elderly (≥65 years)

As metformin is excreted via the kidney, and elderly patients have a tendency to decrease renal function, elderly patients taking VIDASTEAD-M should have their renal function monitored regularly.

Renal impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

The maximum daily dose of metformin should preferably be divided into 2-3 daily doses. Factors that may increase the risk of lactic acidosis (see section 4.4) should be reviewed before considering initiation of metformin in patients with GFR <60 mL/min.

If no adequate strength of VIDASTEAD-M is available, individual mono-components should be used instead of the fixed dose combination.

GFR mL/min	Metformin	Vildagliptin
60-89	Maximum daily dose is 3000 mg. Dose reduction may be considered in relation to declining renal function.	No dose adjustment.
45-59	Maximum daily dose is 2000 mg. The starting dose is at most half of the maximum dose.	Maximal daily dose is 50 mg.
30-44	Maximum daily dose is 1000 mg. The starting dose is at most half of the maximum dose.	
<30	Metformin is contraindicated.	

Hepatic impairment

VIDASTEAD-M should not be used in patients with hepatic impairment, including those with pre-treatment alanine aminotransferase (ALT) or aspartate aminotransferase (AST) >3 times the upper limit of normal (ULN).

Paediatric population

VIDASTEAD-M is not recommended for use in children and adolescents (<18 years). The safety and efficacy of VIDASTEAD-M in children and adolescents (<18 years) have not been established. No data is available.

Method of Administration

Oral use.
Taking VIDASTEAD-M with or just after food may reduce gastrointestinal symptoms associated with metformin.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients in VIDASTEAD-M
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)
- Severely reduced kidney function (GFR <30 mL/min)
- Acute conditions with potential to alter renal function, such as:
 - dehydration
 - severe infection
 - shock
 - intravascular administration of iodinated contrast agents
- Acute or chronic disease which may cause tissue hypoxia, such as:
 - cardiac or respiratory failure
 - recent myocardial infarction
 - shock
- Hepatic impairment
- Acute alcohol intoxication, alcoholism
- Breastfeeding

4.4 Warnings and Precautions

General

VIDASTEAD-M is not a substitute for insulin in patients requiring insulin and should not be used in patients with type 1 diabetes.

Lactic acidosis
Lactic acidosis is a very rare but serious metabolic complication, most often occurs with acute worsening of renal function, or cardiorespiratory illness or sepsis. Metformin accumulation occurs with acute worsening of renal function and increases the risk of lactic acidosis.
In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.
Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.
Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterized by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (≥5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Administration of iodinated contrast agent
Intravascular administration of iodinated contrast agents may lead to contrast-induced nephropathy, resulting in metformin accumulation and increased risk of

lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Renal function

GFR should be assessed before treatment initiation and regularly thereafter (see section 4.2). Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function (see section 4.3).

Concomitant medicinal products that may affect renal function, result in significant haemodynamic change, or inhibit renal transport and increase metformin systemic exposure, should be used with caution.

Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Hepatic impairment

Patients with hepatic impairment, including those with pre-treatment ALT or AST >3x ULN, should not be treated VIDASTEAD-M.

Liver enzyme monitoring

Rare cases of hepatic dysfunction (including hepatitis) have been reported with vildagliptin. In these cases, the patients were generally asymptomatic without clinical sequelae and liver function tests (LFTs) returned to normal after discontinuation of treatment. LFTs should be performed prior to the initiation of treatment with VIDASTEAD-M in order to know the patient's baseline value. Liver function should be monitored during treatment with VIDASTEAD-M at three-month intervals during the first year and periodically thereafter. Patients who develop increased transaminase levels should be monitored with a second liver function evaluation to confirm the finding and be followed up thereafter with frequent LFTs until the abnormality/abnormalities return to normal. Should an increase in AST or in ALT of 3x ULN or greater persist, withdrawal of VIDASTEAD-M therapy is recommended. Patients who develop jaundice or other signs suggestive of liver dysfunction should discontinue VIDASTEAD-M.
Following withdrawal of treatment with VIDASTEAD-M and LFT normalization, treatment with VIDASTEAD-M should not be re-initiated.

Skin disorders

Skin lesions, including blistering and ulceration have been reported with vildagliptin in extremities of monkeys in non-clinical toxicology studies. Although skin lesions were not observed at an increased incidence in clinical trials, there was limited experience in patients with diabetic skin complications. Furthermore, there have been post-marketing reports of bullous and exfoliative skin lesions. Therefore, in keeping with routine care of the diabetic patient, monitoring for skin disorders, such as blistering or ulceration, is recommended.

Acute pancreatitis

Use of vildagliptin has been associated with a risk of developing acute pancreatitis. Patients should be informed of the characteristic symptoms of acute pancreatitis. If pancreatitis is suspected, vildagliptin should be discontinued; if acute pancreatitis is confirmed, vildagliptin should be restarted. Caution should be exercised in patients with a history of acute pancreatitis.

Hypoglycaemia

Sulphonylureas are known to cause hypoglycaemia. Patients receiving vildagliptin in combination with a sulphonylurea may be at risk for hypoglycaemia. Therefore, a lower dose of sulphonylurea may be considered to reduce the risk of hypoglycaemia.

Surgery

Metformin must be discontinued at the time of surgery under general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Severe and disabling arthralgia

There have been post marketing reports of severe and disabling arthralgia in patients taking DPP-4 inhibitors. The time to onset of symptoms following initiation of drug therapy varied from one day to years. Patients experienced relief of symptoms upon discontinuation of the medication. A subset of patients experienced a recurrence of symptoms when restarting the same drug or a different DPP-4 inhibitor. Consider DPP-4 inhibitors as a possible cause for severe joint pain and discontinue drug if appropriate.

4.5 Interactions

There have been no formal interaction studies for VIDASTEAD-M. The following statements reflect the information available on the individual active substances.

Vildagliptin

Vildagliptin has a low potential for interactions with co-administered medicinal products. Since vildagliptin is not a cytochrome P (CYP) 450 enzyme substrate and does not inhibit or induce CYP 450 enzymes, it is not likely to interact with active substances that are substrates, inhibitors or inducers of these enzymes.

Results from clinical trials conducted with the oral antidiabetics pioglitazone, metformin and glyburide in combination with vildagliptin have shown no clinically relevant pharmacokinetic interactions in the target population.

Drug-drug interaction studies with digoxin (P-glycoprotein substrate) and warfarin (CYP2C9 substrate) in healthy subjects have shown no clinically relevant pharmacokinetic interactions after co-administration with vildagliptin.

Drug-drug interaction studies in healthy subjects were conducted with amlodipine, ramipril, valsartan and simvastatin. In these studies, no clinically relevant pharmacokinetic interactions were observed after co-administration with vildagliptin. However, this has not been established in the target population.

Combination with ACE-inhibitors

There may be an increased risk of angioedema in patients concomitantly taking ACE- inhibitors.

As with other oral antidiabetic medicinal products the hypoglycaemic effect of vildagliptin may be reduced by certain active substances, including thiazides, corticosteroids, thyroid products and sympathomimetics.

Metformin

Combinations not recommended

Alcohol

Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in cases of fasting, malnutrition or hepatic impairment.

Iodinated contrast agents

Metformin must be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Combinations requiring precautions for use

Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Glucocorticoids, beta-2-agonists, and diuretics have intrinsic hyperglycaemic activity. The patient should be informed and more frequent blood glucose monitoring performed, especially at the beginning of treatment. If necessary, the dosage of VIDASTEAD-M may need to be adjusted during concomitant therapy and on its discontinuation.

Angiotensin converting enzyme (ACE) inhibitors may decrease the blood glucose levels. If necessary, the dosage of the anti-hyperglycaemic medicinal product should be adjusted during therapy with the other medicinal product and on its discontinuation.

Concomitant use of medicinal products that interfere with common renal tubular transport systems involved in the renal elimination of metformin (e.g. organic cationic transporter-2 [OCT2]/multidrug and toxin extrusion [MATE] inhibitors such as ranolazine, vandetanib, dolutegravir and cimetidine) could increase systemic exposure to metformin.

4.6 Pregnancy, Lactation and Fertility

Pregnancy

There are no adequate data from the use of VIDASTEAD-M in pregnant women. For vildagliptin studies in animals have shown reproductive toxicity at high doses. For metformin, studies in animals have not shown reproductive toxicity. Studies in animals performed with vildagliptin and metformin have not shown evidence of teratogenicity, but foetotoxic effects at maternotoxic doses. The potential risk for humans is unknown. VIDASTEAD-M should not be used during pregnancy.

Breast-feeding

Studies in animals have shown excretion of both metformin and vildagliptin in milk. It is unknown whether vildagliptin is excreted in human milk, but metformin is excreted in human milk in low amounts. Due to both the potential risk of neonate hypoglycaemia related to metformin and the lack of human data with vildagliptin, VIDASTEAD-M should not be used during breast-feeding.

Fertility

No studies on the effect on human fertility have been conducted for VIDASTEAD-M.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Patients who may experience dizziness as an undesirable effect should avoid driving vehicles or using machines.

4.8 Undesirable Effects

There have been no therapeutic clinical trials conducted with VIDASTEAD-M. However, bioequivalence of VIDASTEAD-M with co-administered vildagliptin and metformin has been demonstrated.

The majority of adverse reactions were mild and transient, not requiring treatment discontinuations. No association was found between adverse reactions and age, ethnicity, duration of exposure or daily dose. Vildagliptin use is associated with the risk of development of pancreatitis. Lactic acidosis has been reported following the use of metformin, especially in patients with underlying renal impairment.

Tabulated list of adverse reactions

Adverse reactions reported in patients who received vildagliptin in double-blind studies as monotherapy and add-on therapies are listed below by system organ class and absolute frequency. Frequencies are defined as very common (≥1/10,000; common ≥1/1000 to <1/100; uncommon ≥1/1,000 to <1/100; rare ≥1/10,000 to <1/1,000; very rare (<1/10,000), not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1: Adverse reactions reported in patients who received vildagliptin and metformin (as mono-components or as fixed dose combination), or in combination with other anti-diabetic treatments, in clinical trials and in post-marketing experience

System organ class - adverse reaction	Frequency
Infections and infestations	
Upper respiratory tract infection	Common
Nasopharyngitis	Common
Metabolism and nutrition disorders	
Vitamin B12 decrease/deficiency	Common
Hypoglycaemia	Uncommon
Loss of appetite	Uncommon
Lactic acidosis	Very rare*
Nervous system disorders	
Dizziness	Common
Headache	Common
Tremor	Common
Metallic taste	Uncommon
Gastrointestinal disorders	
Vomiting	Common
Diarrhoea	Common
Nausea	Common
Gastro-oesophageal reflux disease	Common
Flatulence	Common
Constipation	Common
Abdominal pain including upper	Common
Pancreatitis	Uncommon
Hepatobiliary disorders	
Hepatitis	Uncommon
Skin and subcutaneous tissue disorders	
Hyperhidrosis	Common
Pruritis	Common
Rash	Common
Dermatitis	Common
Erythema	Uncommon
Urticaria	Uncommon
Exfoliative and bullous skin lesions, including bullous pemphigoid	Not Known [†]
Cutaneous vasculitis	Not Known [†]
Musculoskeletal and connective tissue disorders	
Arthalgia	Common
Myalgia	Uncommon
General disorders and administration site conditions	
Asthenia	Common
Fatigue	Uncommon
Chills	Uncommon
Oedema peripheral	Uncommon
Investigations	
Abnormal liver function tests	Uncommon
* Adverse reactions reported in patients who received metformin as monotherapy and that were not observed in patients who received vildagliptin+metformin fixed dose combination. Refer to summary of product characteristics for metformin for additional information.	
[†] Based on post-marketing experience.	

Description of selected adverse reactions

Vildagliptin

Hepatic impairment

Rare cases of hepatic dysfunction (including hepatitis) have been reported with vildagliptin. In these cases, the patients were generally asymptomatic without clinical sequelae and liver function returned to normal after discontinuation of treatment. In data from controlled monotherapy and add-on therapy trials of up to 24 weeks in duration, the incidence of ALT or AST elevations ≥3x ULN (classified as present on at least 2 consecutive measurements or at the final on-treatment visit) was 0.2%, 0.3% and 0.2% for vildagliptin 50 mg once daily, vildagliptin 50 mg twice daily and all comparators, respectively. These elevations in transaminases were generally asymptomatic, non-progressive in nature and not associated with cholestasis or jaundice.

Angioedema

Rare cases of angioedema have been reported on vildagliptin at a similar rate to controls. A greater proportion of cases were reported when vildagliptin was administered in combination with an ACE inhibitor. The majority of events were mild in severity and resolved with ongoing vildagliptin treatment.

Hypoglycaemia

Hypoglycaemia was uncommon when vildagliptin (0.4%) was used as monotherapy in comparative controlled monotherapy studies with an active comparator or placebo (0.2%). No severe or serious events of hypoglycaemia were reported. When used as add-on to metformin, hypoglycaemia occurred in 1% of vildagliptin-treated patients and in 0.4% of placebo-treated patients. When pioglitazone was added, hypoglycaemia occurred in 0.6% of vildagliptin-treated patients and in 1.9% of placebo-treated patients. When sulphonylurea was added, hypoglycaemia occurred in 1.2% of vildagliptin treated patients and in 0.6% of placebo-treated patients. When sulphonylurea and metformin were added, hypoglycaemia occurred in 5.1% of vildagliptin-treated patients and in 1.9% of placebo-treated patients. In patients taking vildagliptin in combination with insulin, the incidence of hypoglycaemia was 1.4% for vildagliptin and 1.6% for placebo.

Metformin

Decrease of vitamin B12 absorption

A decrease in vitamin B12 absorption with decrease in serum levels has been observed very rarely in patients who have been treated with metformin over a long period. Consideration of such aetiology is recommended if a patient presents with megaloblastic anaemia.

Liver function

Isolated cases of liver function test abnormalities or hepatitis resolving upon metformin discontinuation have been reported.

Gastrointestinal disorders

Gastrointestinal adverse reactions occur most frequently during initiation of therapy and resolve spontaneously in most cases. To prevent them, it is recommended that metformin be taken in 2 daily doses during or after meals. A slow increase in the dose may also improve gastrointestinal tolerability.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorization of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions.

4.9 Overdose

No data is available with regard to overdose of VIDASTEAD-M.

Vildagliptin

Information regarding overdose with vildagliptin is limited.

Symptoms

Information on the likely symptoms of overdose with vildagliptin was taken from a rising dose tolerability study in healthy subjects given vildagliptin for 10 days. At 400 mg, there were three cases of muscle pain, and individual cases of mild and transient paraesthesia, fever, oedema and a transient increase in lipase levels. At 600 mg, one subject experienced oedema of the feet and hands, and increases in creatine phosphokinase (CPK), AST, C-reactive protein (CRP) and myoglobin levels. Three other subjects experienced oedema of the feet, with paraesthesia in two cases. All symptoms and laboratory abnormalities resolved without treatment after discontinuation of the study medicinal product.

Metformin

A large overdose of metformin (or co-existing risk of lactic acidosis) may lead to lactic acidosis, which is a medical emergency and must be treated in hospital.

Management

The most effective method of removing metformin is haemodialysis. However, vildagliptin cannot be removed by haemodialysis, although the major hydrolysis metabolite (LAY 151) can. Supportive management is recommended.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamics

Pharmacotherapeutic group: Drugs used in diabetes, combinations of oral blood glucose lowering drugs; ATC code: A10BD08

Mechanism of action

VIDASTEAD-M combines two anti-hyperglycaemic agents with complimentary mechanisms of action to improve glycaemic control in patients with type 2 diabetes: vildagliptin, a member of the islet enhancer class, and metformin hydrochloride, a member of the biguanide class.

Vildagliptin, a member of the islet enhancer class, is a potent and selective dipeptidyl- peptidase-4 (DPP-4) inhibitor. Metformin acts primarily by decreasing endogenous hepatic glucose production.

Pharmacodynamic effects

Vildagliptin

Vildagliptin acts primarily by inhibiting DPP-4, the enzyme responsible for the degradation of the incretin hormones GLP-1 (glucagon-like peptide-1) and GIP (glucose-dependent insulinotropic polypeptide).

The administration of vildagliptin results in a rapid and complete inhibition of DPP-4 activity resulting in increased fasting and postprandial endogenous levels of the incretin hormones GLP-1 and GIP.

By increasing the endogenous levels of these incretin hormones, vildagliptin enhances the sensitivity of beta cells to glucose, resulting in improved glucose-dependent insulin secretion. Treatment with vildagliptin 50-100 mg daily in patients with type 2 diabetes significantly improved markers of beta cell function including HOMA-β (Homeostasis Model Assessment-β), proinsulin to insulin ratio and measures of beta cell responsiveness from the frequently- sampled meal tolerance test. In non-diabetic (normal glycaemic) individuals, vildagliptin does not stimulate Insulin secretion or reduce glucose levels.

By increasing endogenous GLP-1 levels, vildagliptin also enhances the sensitivity of alpha cells to glucose, resulting in more glucose-appropriate glucagon secretion. The enhanced increase in the insulin/glucose ratio during hyperglycaemia due to increased incretin hormone levels results in a decrease in fasting and postprandial hepatic glucose production, leading to reduced glycaemia.

The known effect of increased GLP-1 levels delaying gastric emptying is not observed with vildagliptin treatment.

Metformin

Metformin is a biguanide with anti-hyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia or increased weight gain.

Metformin may exert its glucose-lowering effect via three mechanisms:

- by reduction of hepatic glucose production through inhibition of gluconeogenesis and glycogenolysis
- in muscle, by modestly increasing insulin sensitivity, improving peripheral

glucose uptake and utilization

- by delaying intestinal glucose absorption
- Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase and increases the transport capacity of specific types of membrane glucose transporters (GLUT-1 and GLUT-4). In humans, independently of its action on glycaemia, metformin has favourable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies: metformin reduces serum levels of total cholesterol, LDL cholesterol and triglycerides.

The prospective randomized UKPDS (UK Prospective Diabetes Study) study has established the long-term benefit of intensive blood glucose control in type 2 diabetes. Analysis of the results for overweight patients treated with metformin after failure of diet alone showed:

- a significant reduction in the absolute risk of any diabetes-related complication in the metformin group (29.8 events/1000 patient-years) versus diet alone (43.3 events/1000 patient-years), p=0.0023, and versus the combined sulphonylurea and insulin monotherapy groups (40.1 events/1000 patient-years), p=0.0034
- a significant reduction in the absolute risk of diabetes-related mortality: metformin 7.5 events/1000 patient-years, diet alone 12.7 events/1000 patient-years, p=0.017
- a significant reduction in the absolute risk of overall mortality: metformin 13.5 events/1000 patient-years versus diet alone 20.6 events/1000 patient-years (p=0.011), and versus the combined sulphonylurea and insulin monotherapy groups 18.9 events/1000 patient-years (p=0.021)
- a significant reduction in the absolute risk of myocardial infarction: metformin 11 events/1000 patient-years, diet alone 18 events/1000 patient-years (p=0.01)

5.2 Pharmacokinetics

VIDASTEAD-M

Absorption

Bioequivalence has been demonstrated between VIDASTEAD-M at dose strengths (50 mg/850 mg and 50 mg/1000 mg) versus free combination of vildagliptin and metformin hydrochloride tablets at the corresponding doses.

Food does not affect the extent and rate of absorption of vildagliptin from VIDASTEAD-M. The rate and extent of absorption of metformin from VIDASTEAD-M 50 mg/1000 mg was decreased when given with food as reflected by the decrease in C_{max} by 26%, AUC by 7% and delayed T_{max} (2.0 to 4.0 h).

The following statements reflect the pharmacokinetic properties of the individual active substances of VIDASTEAD-M.

Vildagliptin

Absorption

Following oral administration in the fasting state, vildagliptin is rapidly absorbed with peak plasma concentrations observed at 1.7 hours. Food slightly delays the time to peak plasma concentration to 2.5 hours, but does not alter the overall exposure (AUC). Administration of vildagliptin with food resulted in a decreased C_{max} (19%) compared to dosing in the fasting state. However, the magnitude of change is not clinically significant, so that vildagliptin can be given with or without food. The absolute bioavailability is 85%.

Distribution

The plasma protein binding of vildagliptin is low (9.3%) and vildagliptin distributes equally between plasma and red blood cells. The mean volume of distribution of vildagliptin at steady-state after intravenous administration (V_{ss}) is 71 L, suggesting extravascular distribution.

Biotransformation

Metabolism is the major elimination pathway for vildagliptin in humans, accounting for 69% of the dose. The major metabolite (LAY 151) is pharmacologically inactive and is the hydrolysis product of the cyano moiety, accounting for 57% of the dose, followed by the amide hydrolysis product (4% of dose). DPP-4 contributes partially to the hydrolysis of vildagliptin based on an in vivo study using DPP-4 deficient rats. Vildagliptin is not metabolized by CYP 450 enzymes to any quantifiable extent, and accordingly the metabolic clearance of vildagliptin is not anticipated to be affected by co-medications that are CYP 450 inhibitors and/or inducers. In vitro studies demonstrated that vildagliptin does not inhibit/induce CYP 450 enzymes. Therefore, vildagliptin is not likely to affect metabolic clearance of co-medications metabolized by CYP 1A2, CYP 2C8, CYP 2C9, CYP 2C19, CYP 2D6, CYP 2E1 or CYP 3A4/5.

Elimination

Following oral administration of [14C] vildagliptin, approximately 85% of the dose was excreted into the urine and 15% of the dose was recovered in the faeces. Renal excretion of the unchanged vildagliptin accounted for 23% of the dose after oral administration. After intravenous administration to healthy subjects, the total plasma and renal clearances of vildagliptin are 41 and 13 L/h, respectively. The mean elimination half-life after intravenous administration is approximately 2 hours. The elimination half-life after oral administration is approximately 3 hours.

Linearity/non-linearity

The C_{max} for vildagliptin and the area under the plasma concentrations versus time curves (AUC) increased in an approximately dose proportional manner over the therapeutic dose range.

Characteristics in patients

Gender: No clinically relevant differences in the pharmacokinetics of vildagliptin were observed between male and female healthy subjects within a wide range of age and body mass index (BMI). DPP-4 inhibition by vildagliptin is not affected by gender. Age: In healthy elderly subjects (≥ 70 years), the overall exposure of vildagliptin (100 mg once daily) was increased by 32%, with an 18% increase in peak plasma concentration as compared to young healthy subjects (18-40 years). These changes are not considered to be clinically relevant, however. DPP-4 inhibition by vildagliptin is not affected by age.

Hepatic impairment: In subjects with mild, moderate or severe hepatic impairment (Child-Pugh A-C) there were no clinically significant changes (maximum ~30%) in exposure to vildagliptin.

Renal impairment: In subjects with mild, moderate, or severe renal impairment, systemic exposure to vildagliptin was increased (C_{max} 8-66%; AUC 32-134%) and total body clearance was reduced compared to subjects with normal renal function. Ethnic group: Limited data suggests that race does not have any major influence on vildagliptin pharmacokinetics.

Metformin

Absorption

After an oral dose of metformin, the maximum plasma concentration (C_{max}) is achieved after about 2.5 h. Absolute bioavailability of a 500 mg metformin tablet is approximately 50-60% in healthy subjects. After an oral dose, the non-absorbed fraction recovered in faeces was 20-30%.

After oral administration, metformin absorption is saturable and incomplete. It is assumed that the pharmacokinetics of metformin absorption are non-linear. At the usual metformin doses and dosing schedules, steady state plasma concentrations are reached within 24-48 h and are generally less than 1 µg/mL. In controlled clinical trials, maximum metformin plasma levels (C_{max}) did not exceed 4 µg/mL, even at maximum doses.

Food slightly delays and decreases the extent of the absorption of metformin. Following administration of a dose of 850 mg, the plasma peak concentration was 40% lower, AUC was decreased by 25% and time to peak plasma concentration was prolonged by 35 minutes. The clinical relevance of this decrease is unknown.

Distribution

Plasma protein binding is negligible. Metformin partitions into erythrocytes. The mean volume of distribution (V_d) ranged between 63-276 L.

Biotransformation

Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Metformin is eliminated by renal excretion. Renal clearance of metformin is >400 mL/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 h. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Tablet content

Microcrystalline cellulose, Magnesium stearate, Hydroxypropylcellulose, Low substituted HPC, Methylene chloride, Isopropanol

Film-coating

Hypromellose, Titanium dioxide, Iron oxide yellow, Macrogol, Talc

6.2 Incompatibilities

Not applicable

6.3 Shelf life

Please refer to the outer carton

6.4 Storage condition

Store below 30°C
Store in the original package to protect from light and moisture

6.5 Nature and contents of container

Available in Alu-Alu blister pack of 3x10 or 6x10 film-coated tablets
Not all pack sizes may be marketed

6.6 Special precautions for disposal

No special requirements

7. MANUFACTURER

Intas Pharmaceuticals Limited
Plot No. 457, 458 Matoda, Plot No. 191/218P
Chacharwadi Sarkhej Bavla Highway
Ahmedabad, 382 210, India

8. PRODUCT REGISTRATION HOLDER

Lotus Healthcare Malaysia Sdn. Bhd.
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40150 Shah Alam, Malaysia

9. DATE OF REVISION OF THE TEXT

30/10/2024